POWER BI DATA ANALYTICS PROJECT

LOGISTICS VISION

Operations & performance analysis

Load/trip dates: Jan 2022–Dec 2024



PROJECT TEAM

The team behind Logistics Vision

PRESENTED BY

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Which decisions can our logistics data reliably support?

01 Are deliveries meeting scheduled commitments?

02 Which costs and fuel balances need attention?

03 Does asset data measure usable capacity?

04 How do price and fuel changes affect contribution?

Validate service, cost and capacity measures before committing resources

First, we organized the data into five groups

1. Master Data	2. Operations	3. Costs	4. Performance	5. Safety
drivers	loads	fuel_purchases	delivery_events	safety_incidents
trucks	trips	maintenance_records	driver_monthly_metrics	
trailers			truck_utilization_metrics	
customers				
facilities				
routes				

The five groups helped us connect resources, activity, cost and outcomes

OUR ANALYTICAL APPROACH

Our analysis journey

01 Understand the Data

Mapped tables, field meanings and the grain of each dataset.

02 Define the Problem

Framed questions on service, cost, utilisation and safety.

03 Collect the Data

Imported 14 source CSV extracts into Power BI.

04 Clean in Power Query (M)

Set data types, cleaned names and keys, and derived time and service fields.

05 Data Modeling: Star Schema

Connected operational facts with shared business dimensions.

06 Data Processing

Built calculated columns, DAX measures and scenario inputs.

07 Data Analysis

Compared trends, segments and exceptions to locate priorities.

08 Recommendations

Turned findings into actions, owners and measurable KPIs.

Each stage connected a business question to evidence and an action

Verified preparation and data-quality checks

14 CSVs

local source files

Typed fields, headers and standardized driver names

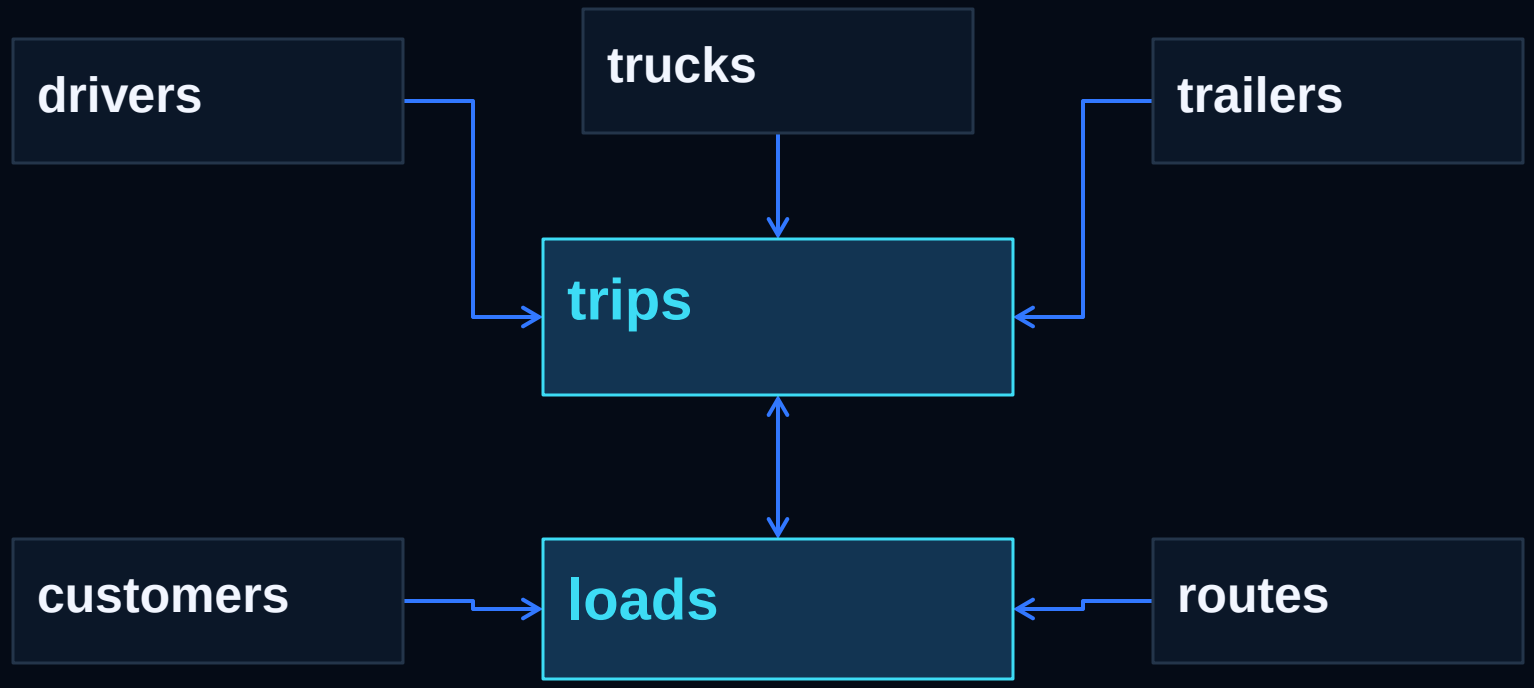
Split date/time fields and rebuilt service flags

Checked keys, duplicates and event chronology

Preserved uncertain records for sensitivity tests

Missing keys and chronology flags require validation; no removal is claimed

Operational facts and shared dimensions



SIMPLIFIED CORE PATHS

16 relationships

14 active, 2 inactive

Other facts cover delivery events, fuel, maintenance, safety and monthly metrics.

Calendar remains disconnected in the saved model.

We used star-schema principles within a model containing multiple fact tables

187 MEASURES AND 2 CALCULATED COLUMNS

KPI logic, denominators and filter context

187

DAX measures

Revenue uses SUMX across the three load revenue components

Gross profit subtracts fuel and maintenance only

Modeled contribution also subtracts damage and assumed wages

Delivery OTD filters Delivery events and timestamp flags

A KPI needs a clear grain, denominator, cost scope and valid filter path

ALL 9 REPORT PAGES COVERED

Nine report pages and the decisions they support

Page	Analytical focus
Home + Company Overview	Navigation and headline KPIs
Operations + Cost & Profitability	Activity, service and cost
Driver + Vehicle Performance	People, capacity and efficiency
Safety + Maps	Risk and destination exposure
WHAT-IF	Pricing and fuel scenarios

The pages connect service, capacity and financial decisions

EXECUTIVE SCORECARD: FULL EMBEDDED DATA

The operating picture in nine measures

$298.62M
Total revenue

$141.23M
Modeled contribution

47.29%
Contribution / revenue

122.16M
Trip miles

85,410
Completed loads

$1.29
Modeled cost per mile

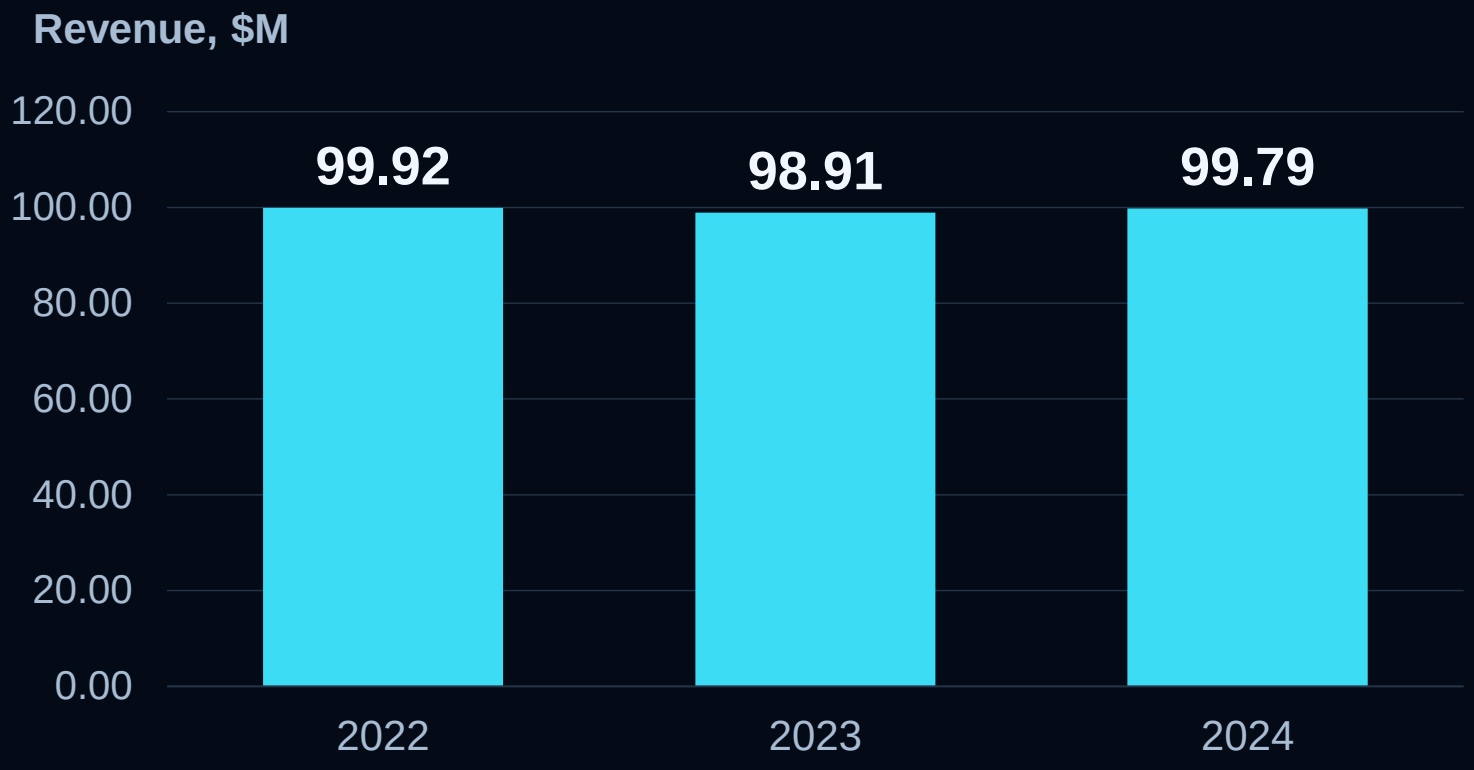
33.28%
Delivery on-time rate

6.45
Trip miles per gallon

1.99
Incidents per 1,000 trips

Contribution excludes unrecorded overhead, insurance and depreciation

Revenue was stable while the mix of work changed



PROBLEM

2024 loads grew 1.74%, revenue grew 0.89%, and revenue per load fell 0.84%. The aggregate requires a mix check.

PROPOSED SOLUTION

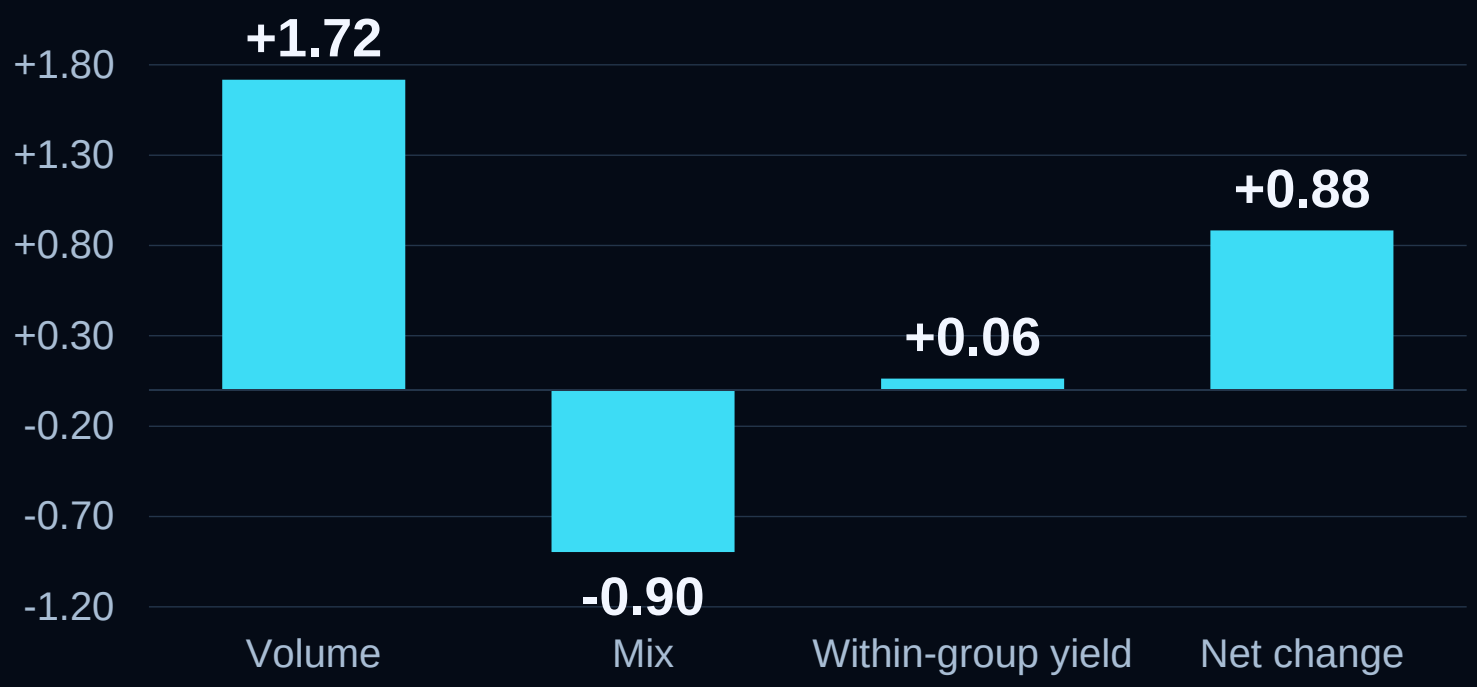
Decompose volume, mix and within-group yield before recommending pricing changes.

Volume recovery did not fully translate into yield growth

DEEP ANALYSIS: YIELD BRIDGE

Mix explains most of the revenue-per-load decline

2023–2024 revenue bridge, $M



PROBLEM

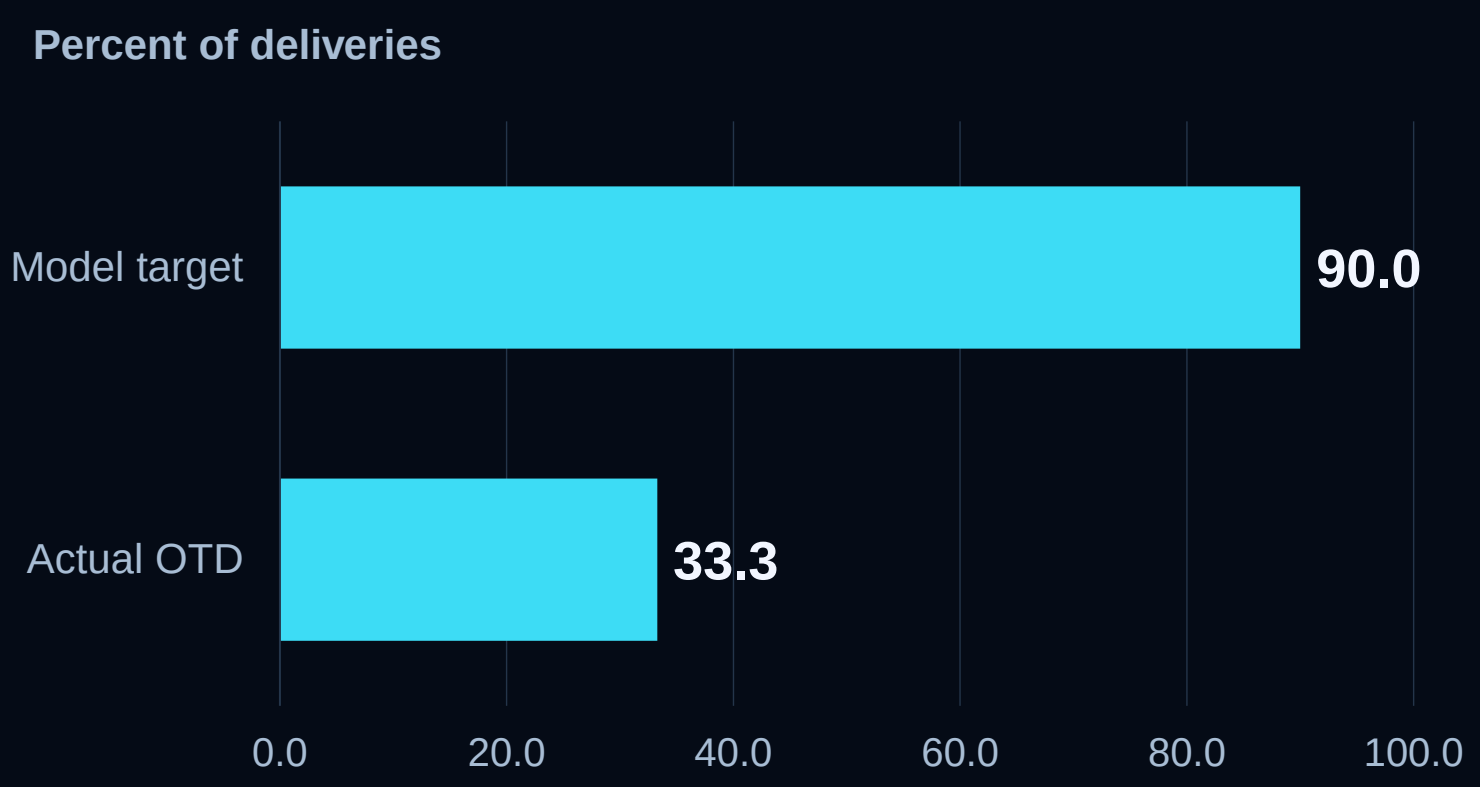
Revenue per load fell 0.84%, while the mix effect was −$0.898M and within-group yield added $0.063M.

PROPOSED SOLUTION

Compare like-for-like segments and distance. Review contract rates before recommending broad repricing.

174 common route/customer-type groups; the components reconcile to +$0.883M

Completed loads concealed a timestamp service gap



PROBLEM

Timestamp OTD is 33.28%, 56.72 points below the 90% model target. Commitment rules need validation.

PROPOSED SOLUTION

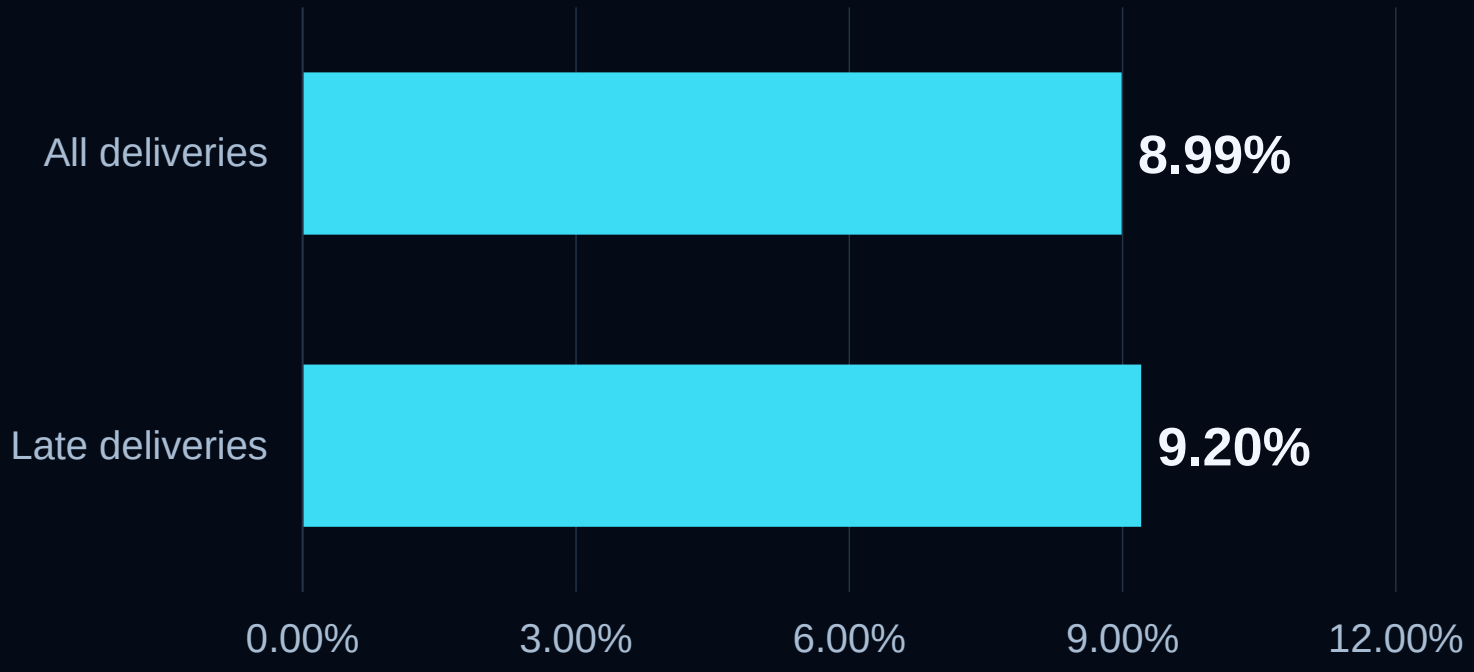
Validate scheduled windows and event chronology, then pilot changes and track OTD plus late-minute severity.

Late deliveries: median 180.05 minutes; P90 324.38 minutes

DEEP ANALYSIS: SERVICE DIAGNOSIS

Late deliveries are spread across the network

Share in the same top five late-count routes



PROBLEM

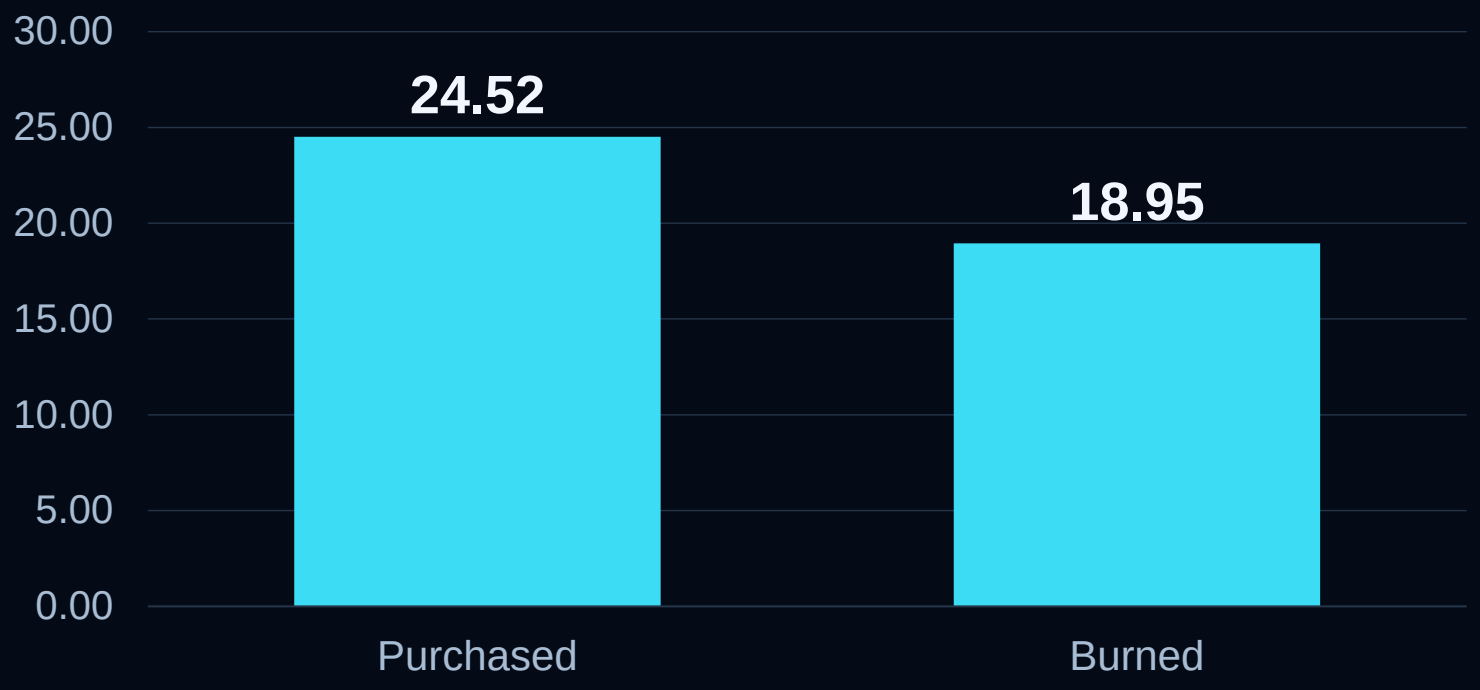
The top five routes contain 9.20% of late deliveries and 8.99% of volume. There is no sharp concentration.

PROPOSED SOLUTION

Validate scheduling rules and event meaning, then pilot changes using reliable stop mappings.

A concentration test does not support a simple “fix a few routes” explanation

Fuel data raised a reconciliation question



PROBLEM

Fuel purchases exceed recorded trip consumption by 5.57M gallons. The gap is not reconciled.

PROPOSED SOLUTION

Reconcile purchases, trip burn and tank inventory by truck and month. Review the unexplained remainder.

The 5.57M-gallon difference is 22.73% of purchases; it is not proven loss

MATCHED FUEL RECONCILIATION

Most of the fuel gap persists in matched truck-months

Purchase date and dispatch date remain different accounting events.

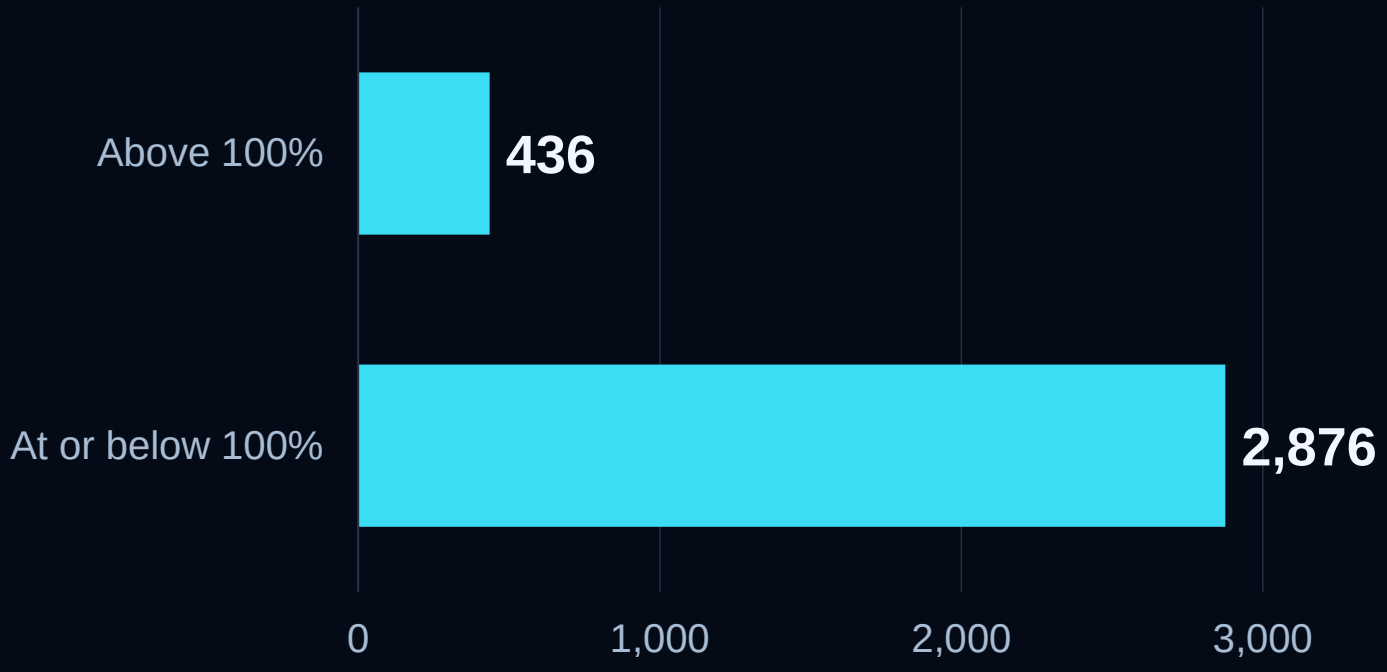
Coverage	Buy, Mgal	Burn, Mgal	Gap, Mgal
Known truck-months Jan 2022–Dec 2024	24.008	18.570	5.437
Unresolved key buckets Jan 2022–Dec 2024	0.486	0.376	0.110
Jan 2025 boundary 201 purchase records	0.025	N/A	N/A

N/A means no comparable burn coverage; opening and closing inventory are still needed

DEEP ANALYSIS: CAPACITY DIAGNOSIS

Validate utilization before making a capacity decision

Source utilization-index records



PROBLEM

436 of 3,312 source index values exceed 100%. All rows match trips divided by calendar days within rounding.

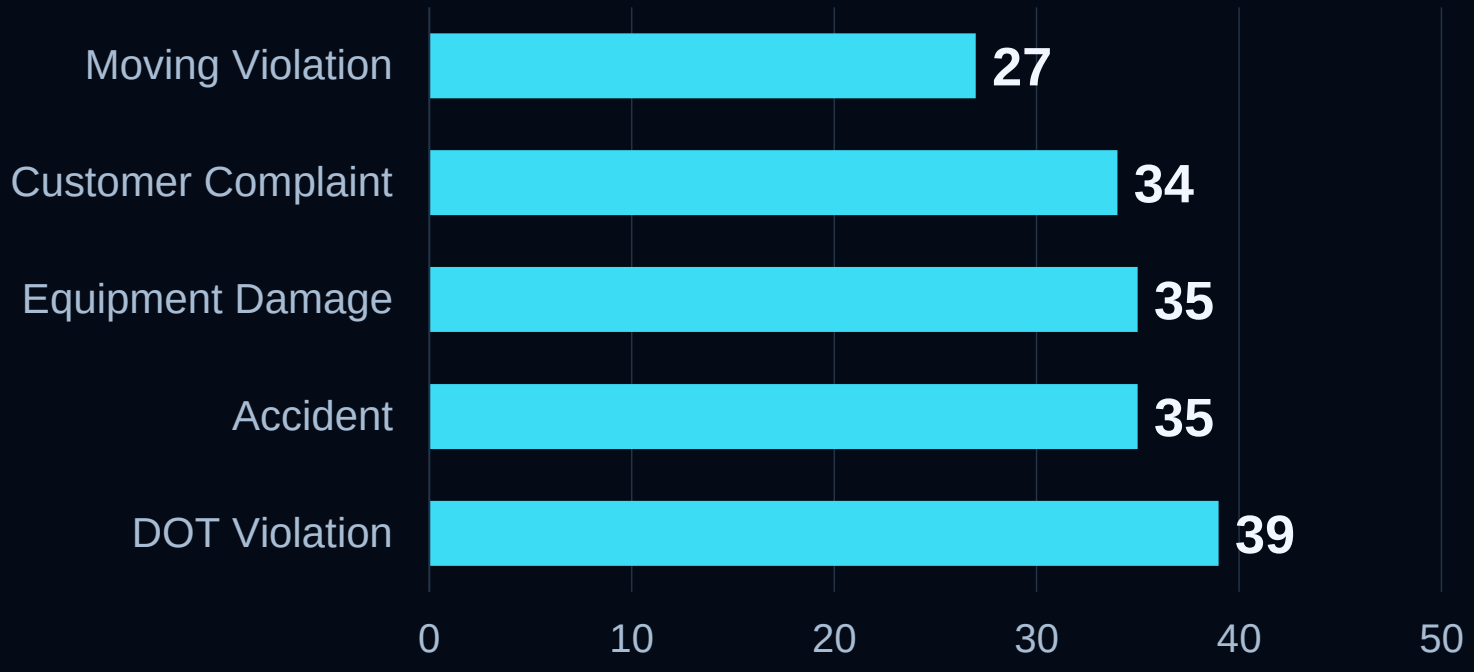
PROPOSED SOLUTION

Confirm the source definition. Build an approved busy-hours / available-hours KPI before changing fleet capacity.

83.04% is a source trip-frequency index, not verified time utilization

Frequency and severity lead to different safety priorities

Recorded incidents



PROBLEM

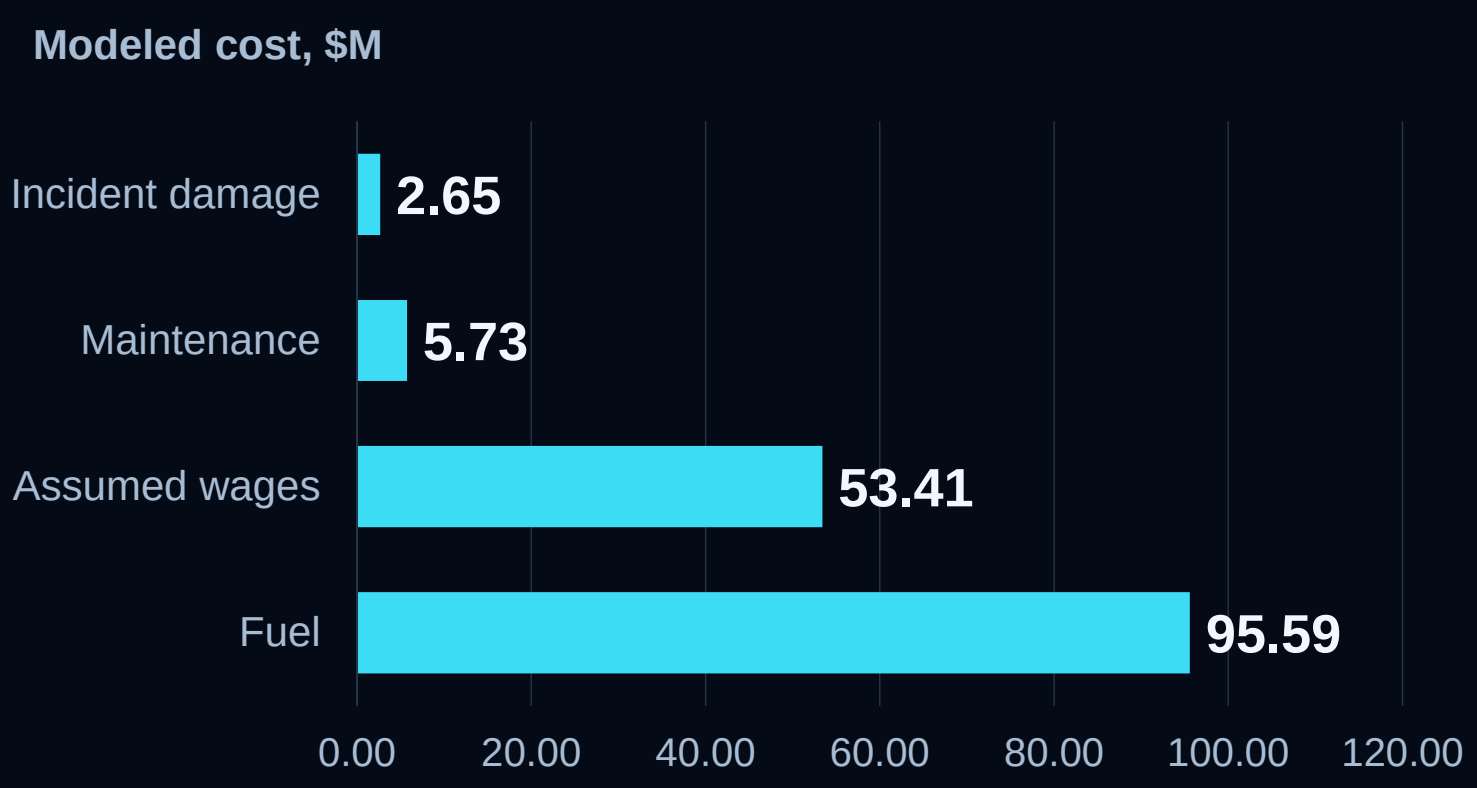
DOT violations lead frequency (39 cases); Equipment Damage leads recorded vehicle and cargo damage cost.

PROPOSED SOLUTION

Review the 64 preventable cases by frequency and severity. Target controls, then track exposure-adjusted outcomes.

Frequency and severity identify different priorities; recoverability is unproven

Modeled contribution depends on the included cost lines



PROBLEM

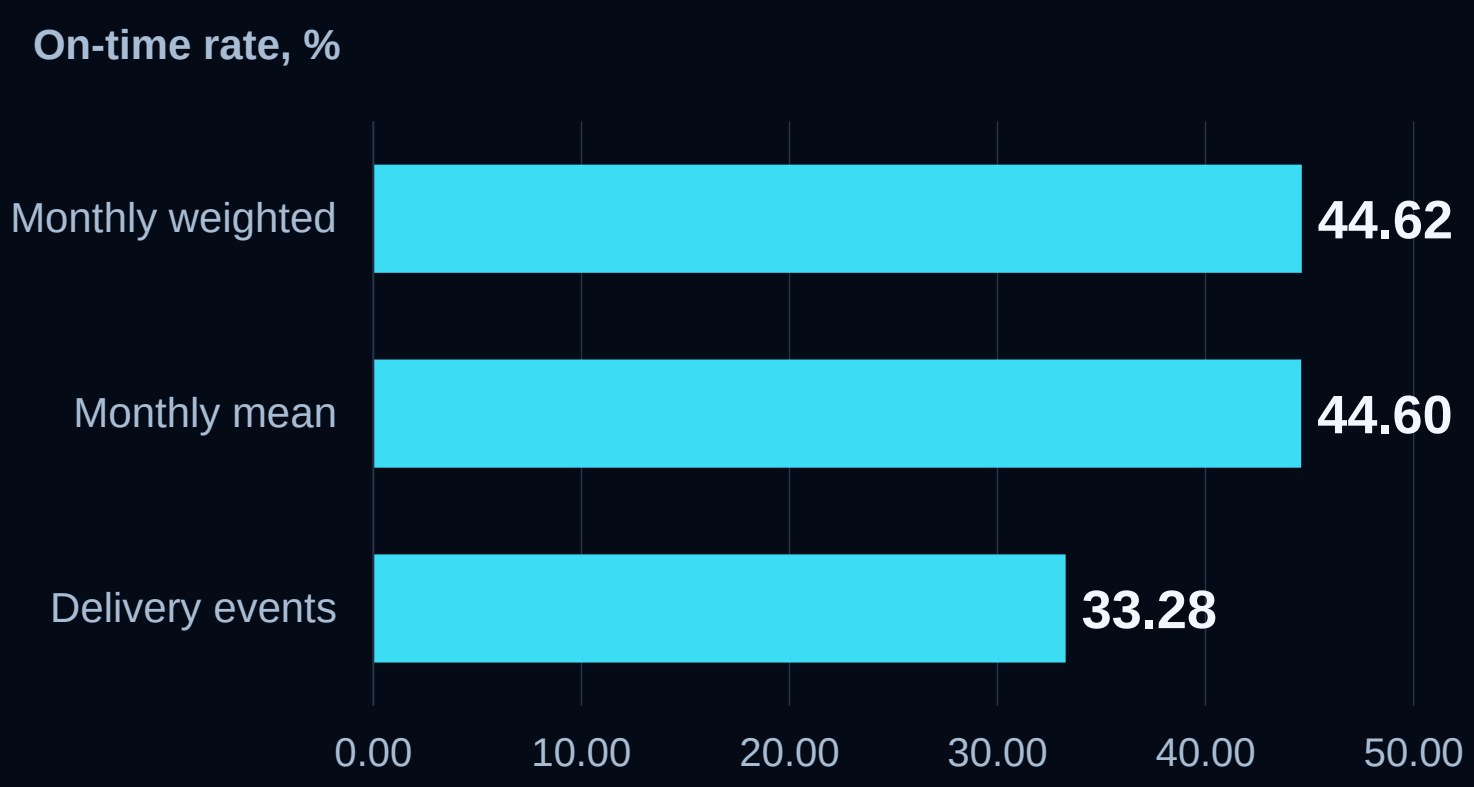
Fuel is 60.74% of modeled cost. The $141.23M contribution excludes unrecorded overhead.

PROPOSED SOLUTION

Reconcile fuel and use actual wages and verified overhead before making full-profitability decisions.

The PBIX measure is named Net Profit; its scope is modeled contribution

Two on-time definitions require source reconciliation



PROBLEM

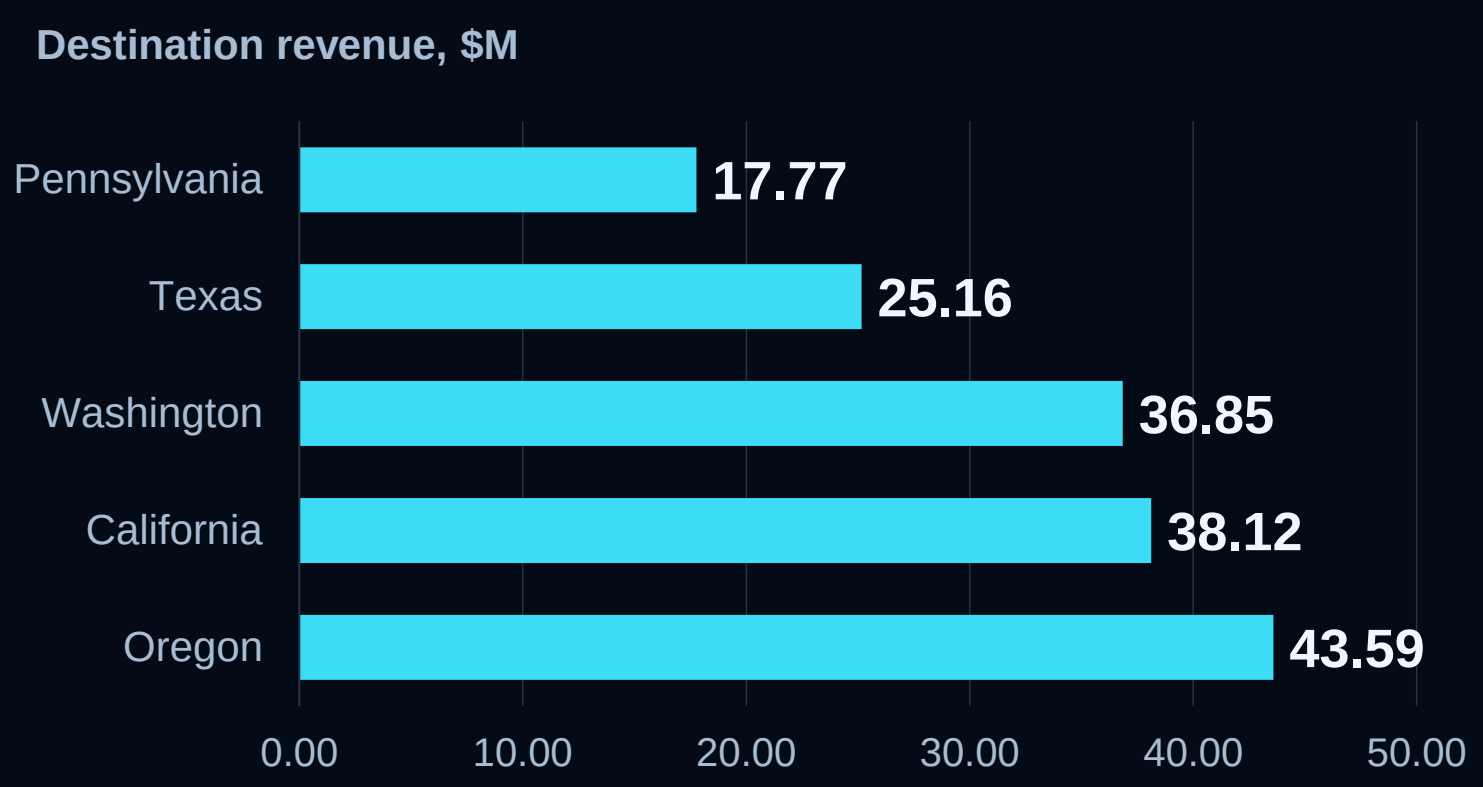
Monthly driver OTD is 44.60% simple or 44.62% trip-weighted, versus 33.28% from delivery events.

PROPOSED SOLUTION

Align cohorts, definitions and driver keys before ranking people or assigning coaching.

Weighting changes the average; it does not reconcile different source definitions

Destination revenue identifies where to investigate



PROBLEM

Three states generate 39.70% of revenue, but destination revenue does not establish lane profit.

PROPOSED SOLUTION

Allocate fuel, labor and maintenance by lane. Compare margin per mile and OTD before repricing.

Geography supports prioritization after cost allocation improves

COST-CONSISTENT SENSITIVITY

At $4.50 fuel, a 5% freight-only uplift is insufficient

Modeled contribution, $M. Baseline $141.23M. Volume, gallons and other costs fixed.

Fuel $/gal	Freight +0%	Freight +5%	Freight +10%
3.50	151.01	164.13	177.26
4.00	138.75	151.88	165.00
4.50	126.49	139.62	152.74

At $4.50: +5% freight gives $139.62M; +5% across all revenue gives $141.42M

The strongest findings change the decision process

33.28%

timestamp-based delivery OTD

1. Service weakness is broad; commitment meaning needs validation

2. Time capacity remains unverified by the source utilization index

3. Mix accounts for most of the revenue-per-load decline

4. Fuel difference persists after truck-month matching

Rank findings by decision significance and evidence, not by the largest number

PROPOSED ACTIONS BASED ON THE FINDINGS

Recommended actions have owners and testable outcomes

P0

model corrections

BI + Operations: validate metric meaning, dates and stop mappings

Fleet + Finance: reconcile fuel balance and unknown keys

Commercial: test matched yield and define the pricing lever

Operations + Safety: pilot supported changes and review cases

Each proposed action has an owner, a baseline and a follow-up KPI

SUGGESTED SEQUENCE AND ACCOUNTABILITY

A staged roadmap separates validation from optimization

6 weeks

suggested initial validation and pilot period

Short term: validate dates, keys and KPI definitions

Medium term: reconcile fuel and pilot supported changes

Long term: add availability data and governed cost allocation

Weekly review: OTD, lateness, fuel variance and yield

Timings are proposed; business owners must approve targets, resources and rollout

Data limitations define the boundary of each conclusion

1,714

unmatched driver trip rows

Original CSV refresh and live visual evaluation unavailable

486 delivery/pickup pairs have actual chronology flags

1,714 trips have unmatched driver keys

Stop mapping, time capacity and cost allocation need validation

Sensitivity tests preserve uncertain records; they do not establish a verified cleaning fix

Lessons for reliable analytics and future development

1 KPI

one certified definition

Date roles must connect before time-filter interpretation

One certified measure per KPI with documented units

Allocated costs for customer and lane profitability

Exposure-adjusted safety and validated scenario tests

A reconciled model enables reliable drill-down analysis

Reliable decisions start with verified analytical meaning

SERVICE

the first operating priority

Verified definitions connect technical work to business decisions

Service, mix and fuel findings identify what to investigate

Comparable scenarios make commercial trade-offs explicit

Controlled pilots can measure whether proposed actions help

The strongest portfolio story includes both insights and validation

Q&A

Which KPI should management address first?

Why do the two on-time rates differ?

How should the WHAT-IF calculation change?

Evidence, definitions and business decisions

Complete dataset inventory

Source table	Rows	Fields
fuel_purchases	196,442	12
delivery_events	170,820	13
loads	85,410	18
trips	85,410	14
driver_monthly_metrics	4,464	10
truck_utilization_metrics	3,312	10
maintenance_records	2,920	13

Source table	Rows	Fields
safety_incidents	170	16
customers	200	8
trailers	180	9
drivers	150	13
trucks	120	11
routes	58	9
facilities	50	11

Business rows span transaction, dimension and monthly summary grains

Relationship and date-filtering audit

Model path	Filter behavior
Customers / routes to loads	Single direction
Loads to trips	Both directions
Drivers / trucks / trailers to trips	Single direction
Trips and safety incidents	1:1, Both directions
Calendar to fact tables	**No relationship**

Calendar filtering requires a model repair

Prepared fields, DAX and scenario inputs

POWER QUERY FEATURES

Total_Revenue combines freight, surcharge and accessorials

Weight_Category and guarded Load_Density describe load mix

real_on_time_flag and delay_minutes compare timestamps

Month, quarter and year fields support date grouping

DAX AND SCENARIO LOGIC

Fleet MPG = trip miles / gallons burned

Modeled cost includes fuel, maintenance, damage and wages

Rate Multiplier is defined from 0.80 to 1.50

Scenario Fuel Price spans $2.00 to $6.00

Prepared fields and DAX measures support analysis across nine report pages

Revenue components and customer segments

REVENUE COMPONENTS

Component	$M	Share
Base freight	262.53	87.91%
Fuel surcharge	29.98	10.04%
Accessorials	6.12	2.05%

CUSTOMER TYPE

Segment	$M	Share
Contract	112.40	37.64%
Spot	94.22	31.55%
Dedicated	92.00	30.81%

Revenue mix describes the book; profitability requires allocated costs

Driver and truck status snapshot

150	Drivers on file	**120**	Trucks on file
124	Active drivers	**92**	Active trucks, 76.67%
26	Terminated drivers, 17.33%	**15**	Trucks in Maintenance status
7.01 h	Mean driver idle-hours metric	**13**	Trucks in Inactive status

76.67% Active status differs from the 83.04% source trip-frequency index

Safety composition and recorded damage

170
Recorded incidents

64
Preventable cases, 37.65%

54
At-fault cases, 31.76%

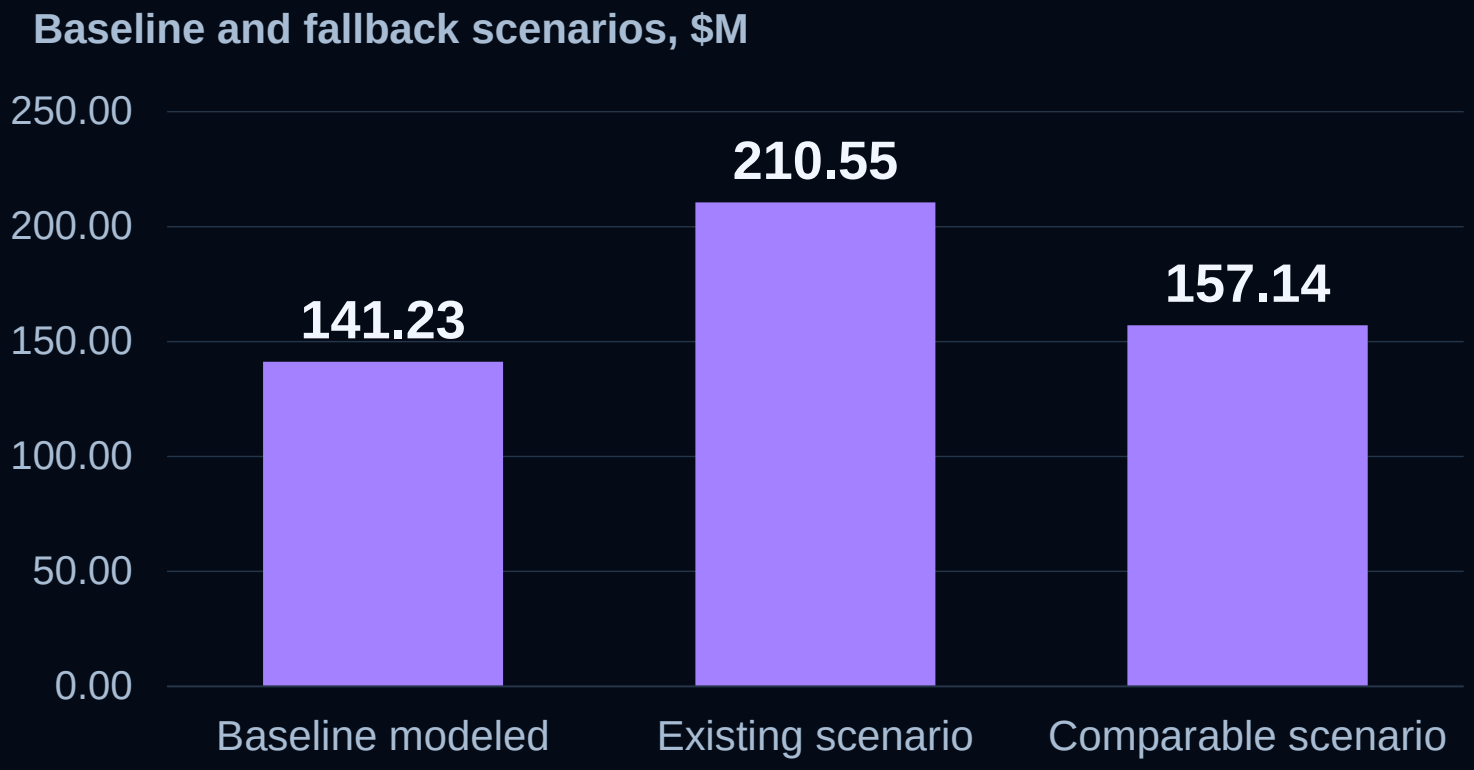
33
Cases involving injury, 19.41%

$2.65M
Recorded vehicle and cargo damage

0.89%
Damage cost / total revenue

The categories can overlap; the counts should not be added together

Scenario audit at fallback inputs



Baseline fuel: $3.8987/gal. Fallback scenario fuel: $3.25/gal.

PROBLEM

Scenario Profit omits $53.41M of wages. Baseline fuel is $3.8987/gal versus $3.25 in fallback scenarios.

PROPOSED SOLUTION

Include wages in both cases. Match fuel price and all baseline inputs, then test for zero contribution delta.

What the current measures can support

85,410

completed loads

Scope: all 85,410 loads and trips are recorded as completed

Mix: Contract customers generate $112.40M revenue

Service: 66.72% of Delivery timestamps are late

Next tests: matched yield, fuel balance and commitment rules

Completion and revenue size describe the dataset; they do not prove profitability

Operational and safety actions with follow-up KPIs

Proposed action	Suggested owner	Follow-up KPI
Weekly late-case review by lane and facility	Operations	Delivery OTD Late minutes
Pilot validated schedule or dispatch changes	Dispatch + Facilities	OTD on pilot lanes Detention minutes
Review 64 preventable cases and target coaching	Safety + Fleet	Preventable cases Incidents / million miles

A case-based pilot can test the causes behind the service gap

Fuel and commercial actions with matched evidence

Proposed action	Suggested owner	Follow-up KPI
Reconcile fuel purchases, trip burn and tank stock	Fleet + Finance	Unexplained variance Gallons per mile
Resolve unmatched driver / truck / trailer keys	Data owners	Unmatched key rate Open exceptions
Compare yield within matched lanes and segments	Commercial + BI	Revenue per load Revenue per mile

Matched records and comparable segments make the next decision clearer

DAX reference: additive measures and units

ACTUAL MODEL FORMULAS

```
Total_Revenue =
SUMX(loads,
  loads[revenue]
  + loads[fuel_surcharge]
  + loads[accessorial_charges])
```

```
Net Profit =
[Total_Revenue] - [Total Cost]
```

UNITS AND COST SCOPE

```
MPG = DIVIDE(
  [Total Miles],
  [Total Gallons Burned], 0)
```

```
Cost Per Mile = DIVIDE(
  [Total Cost], [Total Miles], 0)
```

Net Profit is the saved measure name; the included lines support modeled contribution.

DAX reference: context and comparable scenarios

ACTUAL DELIVERY OTD

```
On_Time_Delivery_Rate =
DIVIDE(
 CALCULATE(COUNTROWS(delivery_events),
  delivery_events[event_type]="Delivery",
  delivery_events[real_on_time_flag]=TRUE()),
 CALCULATE(COUNTROWS(delivery_events),
  delivery_events[event_type]="Delivery"),
 0)
```

SCENARIO COST ALIGNMENT

Current Scenario Profit subtracts:
Fuel + Maintenance + Incident cost

Proposed comparable logic:
[Scenario Revenue]
- [Scenario Fuel Cost]
- [Total Maintenance Cost]
- [Total Incident Cost]
- [Total Driver Wages]

Calendar has no fact link; filter propagation and zero-delta tests remain required.